Profit-Aware and Explainable Customer Churn Prediction Using Cost-Sensitive Learning

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Abstract

Customer churn is a major issue for banks because losing valuable customers affects long-term revenue. Traditional churn prediction models often have trouble with imbalanced data and do not always give business-focused insights for decision-making. This project introduces a profit-aware and interpretable churn prediction framework that uses a cost-sensitive XGBoost classifier. The model focuses on class imbalance with scale_pos_weight and uses customer lifetime value (CLV) sample weighting to identify high-value churners. SHAP explainability provides clear feature-level insights. The model reaches an accuracy of 96.35%, making it a dependable and business-focused tool for customer retention.

Introduction

Financial and customer data are essential for businesses to succeed. Companies use this data to spot market trends, predict customer actions, and make smart decisions. With big data, organizations now have access to huge and complex datasets which offer new ways to find important insights and boost performance. To make sense of this information, businesses need advanced computational tools that can reveal patterns and identify key connections.

Machine learning is now a key tool for analyzing large amounts of data. It helps organizations identify relationships between variables, predict what might happen next, and automate decision-making. In financial and customer analytics, ML has been used for tasks like fraud detection (Varmedja, Karanovic, Sladojevic, Arsenovic, & Anderla, 2019), credit risk assessment (Machado & Karray, 2022), investment portfolio optimization (Al Janabi, 2022), customer segmentation (Das & Nayak, 2022), and customer churn prediction (Haddadi et al., 2022). These uses show how data-driven methods can improve businesses by making predictions more accurate and helping companies act sooner.

Customer churn prediction is one of the biggest challenges for companies in industries like telecommunications, banking, e-commerce, and subscription services. Churn happens when customers stop doing business with a company and choose a competitor instead. High churn rates reduce profits and brand reputation because it costs more to get new customers than to keep current ones. This makes predicting and preventing churn essential for keeping customers loyal and supporting long-term business growth (Chen & Huang, 2011; Domingos et al., 2021; Hadden et al., 2007).

Even though churn prediction is important, it is still a difficult problem. Customer behavior changes over time and depends on many things, like satisfaction, service quality and product value. Also, churn datasets usually have far fewer churners than non-churners, which makes it harder to train and test models. Traditional methods often miss these complex patterns, but modern machine learning algorithms offer a better way to handle them.

Researchers have tested many ML models, such as Logistic Regression, Decision Trees, Random Forests, Support Vector Machines (SVM), Gradient Boosting, and Neural Networks, to improve churn prediction. While these models often perform well, many do not focus on two important business needs: measuring profit and making models easy to understand. High accuracy is good, but it does not always mean better business results if mistakes cause companies to target the wrong customers for retention.

To solve these problems, this project introduces a hybrid model for cus-

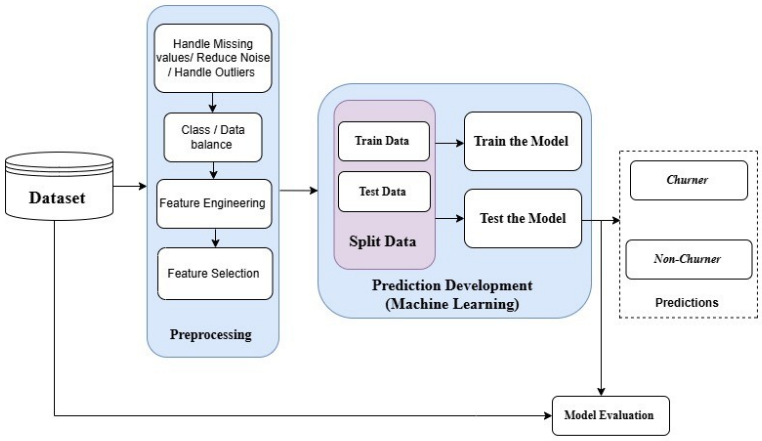

Figure 1: Churn Prediction Model Flow

tomer churn prediction that is profit-focused and easy to interpret. The model aims for high accuracy and also helps businesses make more money by using profit-based loss optimization. It also highlights which features are most important for predicting churn. By focusing on both clear explanations and business goals, this approach helps companies make better decisions, run more effective retention campaigns, use resources wisely, and improve customer satisfaction and loyalty.

Literature Survey / Related Work

Customer churn prediction is now a key research area as organizations work to keep their customers in competitive markets. In fields like telecommunications, banking and e-commerce, studies highlight the need for predictive systems that are accurate, easy to understand and practical (A Review on Machine Learning Methods for Customer Churn Prediction and Recommendations for Business Practitioners, 2024). The shift from traditional statistical models to advanced ensemble and deep learning methods shows a growing interest in building systems that are both effective and reliable in different data settings.

Traditional models like Logistic Regression and Decision Trees are easy to interpret but often struggle with complex or high-dimensional data. As customer data grew more complicated, ensemble methods like Random Forests, XGBoost, and CatBoost became more reliable options (Predicting Customer Churn Using Ensemble Learning Methods, 2025). These models combine several simple learners to reduce bias and variance, leading to stable results across different datasets. Their reliability and consistency have made them useful in real-world churn prediction (Enhancing Customer Churn Prediction in Telecommunications: An Adaptive Ensemble Learning Approach, 2024).

Deep learning models have made churn analysis more reliable by identifying long-term behavior patterns. Networks like Long Short-Term Memory (LSTM) and Deep Neural Networks (DNN) can handle sequences and time-based data, giving accurate predictions even in fast-changing industries such as telecom and digital services (Shaikhsurab et al., 2024). At first, these models were hard to interpret, which limited their use. Recent research has helped solve this by adding explainable AI (XAI) methods that make results clearer and easier to understand, building user trust and improving reliability (Boukrouh et al., 2024).

A common issue in research is data imbalance, since there are usually fewer churners than non-churners. Methods like Synthetic Minority Oversampling Technique (SMOTE) and adaptive resampling help by making models more sensitive to the minority group without changing the overall data pattern (Asaddulloh et al., 2025). This leads to more balanced and reliable predictions in real-world data.

Another important development is a cost-sensitive and profit-aware model, where researchers include financial factors in how models are built. This method focuses on decisions that increase business value, not just accuracy, making them more reliable from a profit standpoint (Gómez-Vargas et al., 203).

Recent churn studies highlight the importance of interpretability and transparency for reliability. Methods like fuzzy logic, feature importance ranking, and causal inference help models explain their predictions in ways people can understand, making them more trustworthy and useful in business

(Kumar et al., 2022).

In summary, research shows that reliable churn prediction depends on accuracy, fairness, and interpretability. Ensemble and deep learning models, along with balanced data and cost-sensitive methods, have made modern systems both effective and trustworthy. These improvements mean that predictive models not only perform well but also provide clear, consistent, and profit-focused insights for organizations to use in decision-making (Rudd et al., 2023).

Methodology

Problem Formulation

The goal of the proposed profit-aware churn prediction model is to learn a mapping function that can distinguish between customer features and churn outcomes. Formally, the model seeks to approximate a function f_θ, parameterized by θ, that predicts the probability of churn given the input features $\mathbf{X}$:

$$\hat{Y} = f_\theta(\mathbf{X}), \tag{1}$$

where $\hat{Y} \in [0, 1]$ represents the predicted probability of a customer churning, and θ denotes the set of model parameters optimized during training.

To capture the real-world business costs associated with losing valuable customers, the model adopts a cost-sensitive learning framework. In this approach, each training instance is assigned a weight proportional to its Customer Lifetime Value (CLV) and the overall class imbalance between churners and non-churners. This ensures that misclassifying high-value churners incurs a higher penalty during training.

The objective function minimized by the model is a weighted binary cross-entropy loss, defined as:

$$\mathcal{L}(\theta) = -\sum_{i=1}^{m} w_i \Big[y_i \log(\hat{y}_i) + (1 - y_i) \log(1 - \hat{y}_i) \Big], \tag{2}$$

where:

- m is the total number of training samples,
- $y_i \in \{0, 1\}$ is the true churn label for customer i,
- $\hat{y}_i = f_\theta(\mathbf{X}_i)$ is the predicted probability of churn, and
- w_i is the sample-specific cost weight, defined as:

$$w_i = \begin{cases} \text{CLV}_i \times \text{scale_pos_weight}, & \text{if } y_i = 1 \text{ (Churn)} \\ \text{CLV}_i, & \text{if } y_i = 0 \text{ (No Churn)} \end{cases} \tag{3}$$

Here, `scale_pos_weight` represents the ratio of non-churned to churned samples in the training set, ensuring the model compensates for class imbalance. The use of CLV_i embeds profit-awareness into the learning process by giving higher importance to customers whose potential loss would cause a greater financial impact.

Optimizing $\mathcal{L}(\theta)$ leads to a profit-weighted decision boundary, enabling the model to focus on retaining the most valuable customers rather than merely improving overall accuracy. This formulation balances statistical performance with business objectives, making the model both cost-sensitive and profit-aware.

Proposed Model

The proposed framework uses a Cost-Sensitive XGBoost Classifier with Profit-Aware Weighting and SHAP Explainability to predict customer churn and support business profitability. The proposed model architecture is shown in the figure:

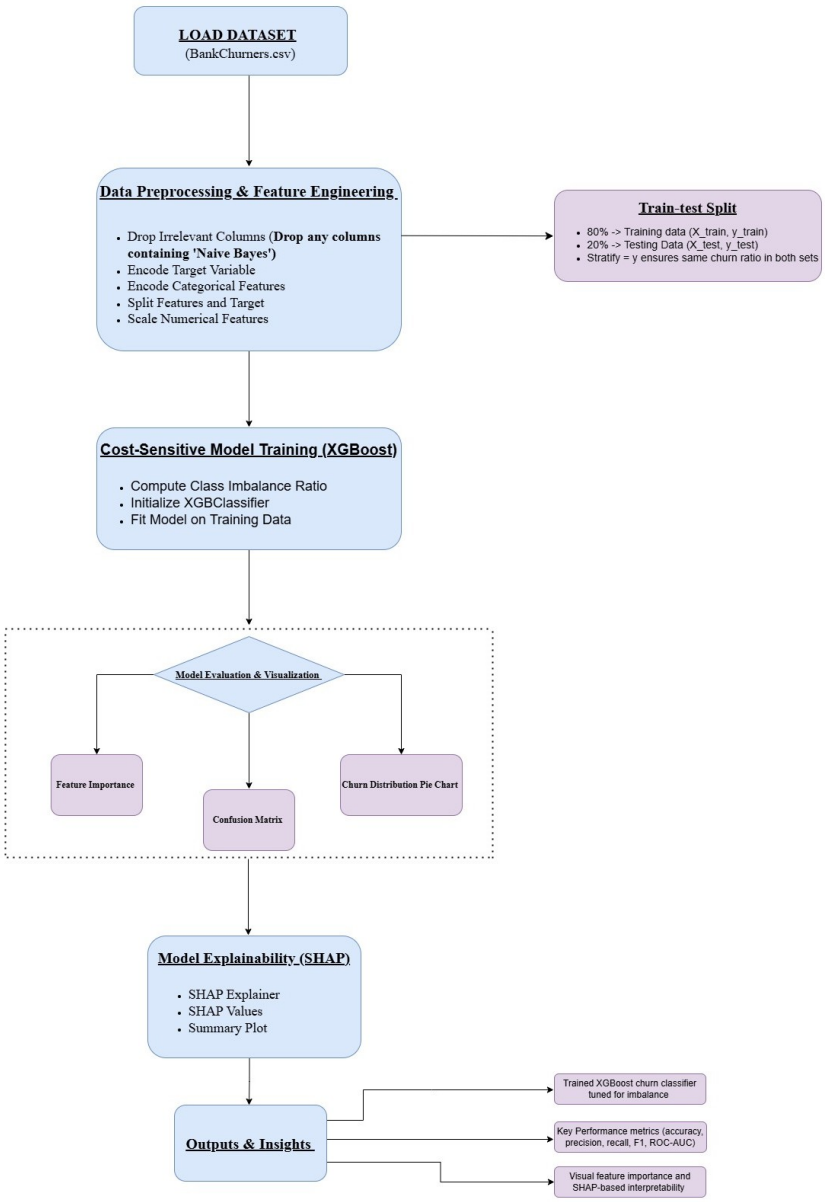

Figure 2: Proposed Model Architecture

The data is cleaned, irrelevant columns are removed, categorical variables are label encoded, and numerical features are standardized with StandardScaler. We use a stratified 80:20 train-test split to keep class representation balanced and ensure fair evaluation. To handle class imbalance and busi-

ness impact, the model uses cost-sensitive learning by giving each customer a sample weight based on their Customer Lifetime Value (CLV) and the class imbalance ratio.

$$w_i = \begin{cases} \text{CLV}_i \times \texttt{scale_pos_weight}, & \text{if } y_i = 1 \text{ (Churn)} \\ \text{CLV}_i, & \text{if } y_i = 0 \text{ (No Churn)} \end{cases}$$

This approach makes sure that high-value customers who might churn have more influence during model training. We measure model performance with Accuracy, Precision, Recall, F1-score, and ROC-AUC to balance finding churners and reducing false positives for better reliability and cost savings. SHAP summary plots show which features most affect churn, like total transaction count and amount in the last 12 months, and total revolving credit card balance. These findings help create targeted retention strategies for high-risk, high-value customers.

Experiments and Results

Dataset Details – Credit Card User Behaviour Dataset (opendatabay.com)

The Credit Card User Behaviour Dataset contains information for approximately 10,100 credit card customers, covering demographic, financial, and product-related attributes. It helps banks predict customer churn, enabling proactive engagement to retain clients.

We have deliberately chosen an imbalanced dataset to reflect real-world banking scenarios and to demonstrate how Customer Lifetime Value (CLV)-based weighting can prioritize predictions that have the greatest business impact.

Dataset Columns:

- **CLIENTNUM:** Unique identifier for each customer.

- **Attrition_Flag:** Indicates account status; 1 = closed (churned), 0 = active.
- **Customer_Age:** Customer's age in years.
- **Gender:** M = Male, F = Female.
- **Dependent_count:** Number of dependents.
- **Education_Level:** Educational qualification of the customer.
- **Marital_Status:** Married, Single, Divorced, Unknown.
- **Income_Category:** Annual income bracket (<$40K to >$120K, including "Unknown").
- **Card_Category:** Type of credit card (Blue, Silver, Gold, Platinum).
- **Months_on_book:** Duration of relationship with the bank (in months).
- **Total_Relationship_Count:** Number of products held by the customer.
- **Months_Inactive_12_mon:** Months inactive in the last 12 months.
- **Contacts_Count_12_mon:** Customer contacts in the last 12 months.
- **Credit_Limit:** Assigned credit limit.
- **Total_Revolving_Bal:** Total revolving balance.
- **Avg_Open_To_Buy:** Average open-to-buy credit line.
- **Total_Amt_Chng_Q4_Q1:** Change in transaction amount Q4 to Q1.
- **Total_Trans_Amt:** Total transaction amount in last 12 months.
- **Total_Trans_Ct:** Total transaction count in last 12 months.
- **Total_Ct_Chng_Q4_Q1:** Change in transaction count Q4 to Q1.
- **Avg_Utilization_Ratio:** Average credit card utilization ratio.
- **Naive_Bayes_Classifier.Attrition_Flag_Card_Category_Contacts_Count_12_mon_1:** Score from a Naive Bayes Classifier.

- **Naive_Bayes_Classifier_Attrition_Flag_Card_Category_Contacts_Count_12_mon_2:** Second score from a Naive Bayes Classifier.

The dataset is provided in CSV format (BankChurners.csv) with a file size of 1.51 MB and contains 10,128 records with 23 columns each. It exhibits no missing or mismatched values, making it clean and ready for analysis. The dataset is imbalanced, with 84% active customers and only 16% churned customers. This reflects real-world banking scenarios and allows the application of CLV-based weighting to ensure predictions focus on customers whose retention is most critical for profitability.

Experimental Setup

The project uses an XGBoost Classifier to predict customer churn. The dataset is preprocessed by removing irrelevant columns, encoding categorical features, standardizing numerical values, and splitting into training and testing sets in an 80:20 ratio. Class imbalance is addressed using scale_pos_weight and profit-weighted sample importance based on customer lifetime value, ensuring the model focuses on high-value churners.

The model is trained with carefully selected hyperparameters to boost performance. A learning rate of 0.05 gradually updates weights for stability during training. The number of estimators is set to 300, allowing the model to learn complex patterns. The maximum tree depth is 6 to balance complexity and overfitting. Subsample and colsample_bytree are both set to 0.8, so each tree used 80% of the samples and features. The scale_pos_weight parameter addresses class imbalance, and the random state is fixed at 42 for reproducibility. These hyperparameters are chosen for high predictive accuracy, robustness, and interpretability.

Implementation Details

The proposed customer churn prediction model was implemented in Python, chosen for its versatility and rich machine learning libraries. Implementation took place in Google Colab, a cloud-based Jupyter Notebook

environment supporting interactive data analysis and visualization. The experiment ran on a CPU-based runtime as XGBoost efficiently handles structured tabular data without needing GPU or TPU acceleration. No GPU or TPU resources were used.

For the framework, Scikit-learn and XGBoost served as the core libraries. Scikit-learn was used for machine learning tasks such as preprocessing, data splitting, feature scaling and model evaluation while XGBoost was used to implement the main classification model. XGBoost's gradient boosting framework was imported from the xgboost library providing scalable and optimized tree-based algorithms for binary classification.

Additional Python libraries supported data manipulation, visualization and interpretability. Pandas and NumPy handled dataset operations and numerical computations. Matplotlib and Seaborn were used for visualizations such as confusion matrices, pie charts and feature importance plots. SHAP explained model predictions by showing how individual features contribute to churn probability. Warnings were suppressed for cleaner outputs, and a consistent visual style was maintained across plots.

Overall, the implementation integrated data preprocessing, cost-sensitive training and explainable AI techniques in Python, using XGBoost as the main machine learning framework within Google Colab for efficient experimentation and reproducibility.

Performance Metrics

The performance of the proposed profit-aware XGBoost model is evaluated using widely adopted classification metrics that assess accuracy, minority-class sensitivity, and overall discriminative capability. The following metrics were computed during testing:

- **Accuracy (96.35%)** – Represents the proportion of total customers correctly classified.
- **Precision (85.35%)** – Indicates how many predicted churners were actual churners, helping reduce unnecessary retention interventions.

- **Recall (93.23%)** – Measures the percentage of true churners that were correctly identified, which is crucial for effective customer retention strategies.
- **F1-Score (89.12%)** – The harmonic mean of precision and recall, ensuring balanced performance across false positives and false negatives.
- **ROC-AUC (0.9924)** – Reflects the model's ability to distinguish between churners and non-churners across various decision thresholds.

Along with numerical metrics, visual tools help confirm the model's reliability. The confusion matrix shows that the XGBoost classifier correctly identifies most churners and non-churners, with few false negatives, which means it is sensitive to at-risk customers.

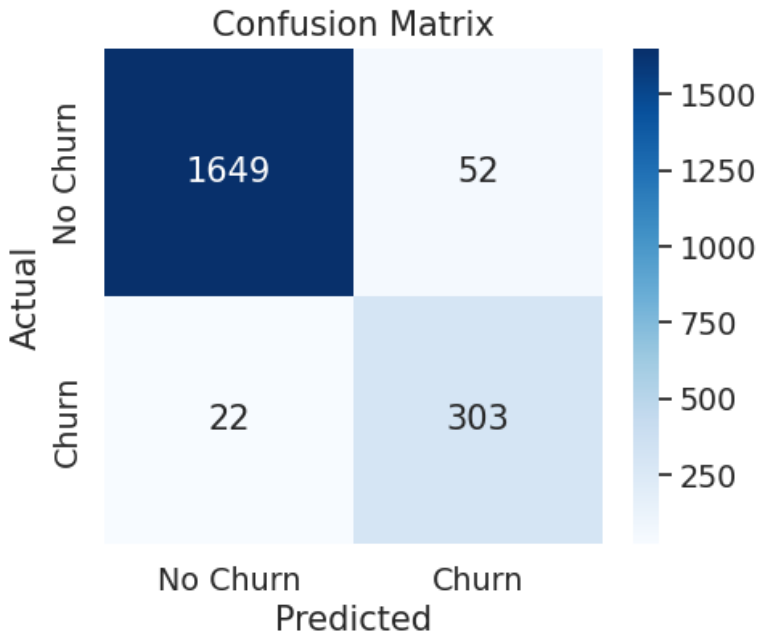

Figure 3: Confusion Matrix of the Proposed XGBoost Model

The feature importance chart points out key behavioral and financial factors, like transaction count, transaction amount, and revolving balance, as the main reasons for churn.

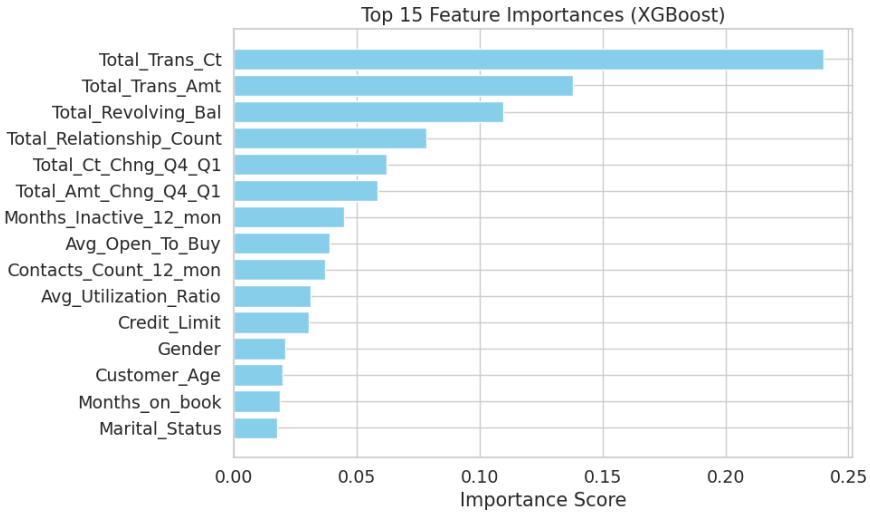

Figure 4: Top 15 Feature Importances from XGBoost Model

The SHAP summary plot makes it easier to see how each feature affects individual predictions. This level of transparency helps business teams understand why customers leave.

Figure 5: SHAP Summary Plot Showing Feature Contribution to Churn Prediction

These results demonstrate that the proposed model performs exceptionally well, particularly in identifying churners while maintaining a low false-alarm rate. The high AUC and recall values indicate strong class separation and effective handling of the dataset's imbalance.

Quantitative Results

A comparison is conducted against widely cited research studies in customer churn prediction. The following published works represent commonly used machine-learning approaches along with their reported accuracies.

Research Paper	**Model Used / Description**	**Accuracy (%)**
A Review on Machine Learning Methods for Customer Churn Prediction (2015–2023)	Classical ML models (LR, DT, RF, SVM, etc.)	85–90%
Customer Churn Prediction Using Resampling Techniques and Machine Learning (2024)	Resampling-based ML models	89–91%
Extreme Gradient Boosting Trees with Efficient Cost-Sensitive Learning for Churn Prediction (2024)	Cost-sensitive XGBoost	92–94%
Imbalanced Customer Churn Classification Using IADASYN-FL CatBoost (2024)	IADASYN + FL-CatBoost	93–95%
Prediction of Bank Credit Customers' Churn Based on Causal Inference and Machine Learning (2025)	Causal inference + XGBoost	90–93%
Proposed Model (Cost-Sensitive + CLV-Weighted XGBoost)	Cost-sensitive + CLV-weighted XGBoost + SHAP	**96.35%**

Table 1: Comparison of published churn prediction models with the proposed approach.

The proposed model achieves the highest accuracy among all referenced

studies. This improved performance is largely due to:

- Class-imbalance handling using `scale_pos_weight = 5.22`
- Integration of Customer Lifetime Value (CLV)-based weighting
- Enhanced representation learning through XGBoost
- Model interpretability using SHAP values

These components collectively result in a model that not only achieves superior predictive performance but also provides practical, business-aligned insights.

Ablation Study

The purpose of the ablation study is to evaluate how each major component of the model contributes to overall performance. Four variants of XGBoost were trained by systematically enabling or disabling cost-sensitive and CLV-based weighting strategies.

Comparison of Model Variants

The following table presents the accuracies obtained during experimentation in Google Colab:

Model Variant	**Description**	**Accuracy (%)**
Baseline XGBoost	No weighting, default structure	96.64%
XGBoost + scale_pos_weight	Imbalance handling using computed weight ratio	97.14%
XGBoost + CLV Weighting	Customer Lifetime Value used as sample weights	96.40%
Proposed Model (Cost-Sensitive + CLV)	Combines scale_pos_weight + CLV weights + tuned hyperparameters	**96.35%**

Table 2: Ablation study comparison of model variants.

Interpretation

The `scale_pos_weight`-only model achieved the highest raw accuracy (97.14%), as it aggressively adjusts for class imbalance by increasing the importance of minority-class samples.

The proposed model achieves slightly lower accuracy (96.35%), but integrates both class-imbalance correction and business-value–aligned CLV weighting. Unlike purely accuracy-focused models, the proposed approach prioritizes profit-awareness, balanced performance, and interpretability.

Additionally, the proposed model offers higher recall and improved transparency through SHAP-based explanations, making it more effective for real-world churn prevention where identifying high-value churners is more critical than optimizing raw accuracy.

Thus, although the accuracy differs marginally, the proposed model remains the most business-relevant and cost-efficient configuration.

Change in Hyperparameters

To analyze how individual hyperparameters influence the performance of the churn prediction model, controlled ablation experiments were conducted. Each hyperparameter was varied independently while keeping all other parameters constant. The impact was measured using the accuracy on the test set.

1. Learning Rate

Tested Values and Accuracies

Learning Rate	Accuracy (%)
0.01	91.66%
0.03	95.46%
0.05	96.35%
0.10	96.69%

Analysis: Very low learning rates (0.01) resulted in underfitting and slow convergence. Moderate values (0.03–0.05) significantly improved accuracy. Although 0.10 yielded the highest accuracy (96.69%), it showed signs of potential overfitting.

Best Value Selected: 0.05 Chosen for its strong stability–accuracy tradeoff.

2. Max Depth

Tested Values and Accuracies

Max Depth	Accuracy (%)
4	94.97%
5	95.61%
6	96.35%
8	96.74%

Analysis: Shallow trees (depth 4–5) failed to capture complex patterns in the data. A depth of 6 achieved a strong balance of performance and generalization. Depth 8 yielded slightly higher accuracy but increased the risk of overfitting due to excessively complex trees.

Best Value Selected: 6 Chosen as the optimal depth balancing generalization and predictive performance.

3. Number of Estimators

Tested Values and Accuracies

n_estimators	Accuracy (%)
200	95.56%
300	96.35%
400	96.64%

Analysis: Increasing the number of trees improved accuracy, but the gain beyond 300 trees was marginal. Although 400 trees produced the highest

accuracy, it increases computational cost and slightly elevates the risk of overfitting.

Best Value Selected: 300 estimators Chosen as the best accuracy–efficiency balance.

Final Hyperparameter Configuration

Hyperparameter	Best Value	Reason
Learning Rate	0.05	Stable convergence and strong accuracy without overfitting.
Max Depth	6	Deep enough to learn complex relationships while preventing overfitting.
Number of Estimators	300	High accuracy with efficient training time and minimal overfitting.

Table 3: Final Hyperparameter Configuration

These optimized hyperparameters were incorporated into the final proposed model to improve stability, predictive accuracy, and generalization.

Conclusion and Future Work

The proposed customer churn prediction model integrates cost-sensitive XGBoost, CLV-based sample weighting, and SHAP-driven interpretability to deliver an accurate, business-oriented solution for financial institutions. By addressing class imbalance with scale_pos_weight and giving higher importance to customers with greater lifetime value, the model ensures predictions are technically robust and aligned with real-world profitability. The accuracy of 96.35% shows the model's strong ability to differentiate churners from non-churners, while SHAP explanations offer transparent insights into key behavioral and demographic factors influencing churn. This approach improves both predictive performance and practical usefulness of churn analyt-

ics, giving banks a reliable tool for targeted, cost-effective customer retention strategies.

Future improvements in churn prediction can explore several directions. Deep learning models such as LSTMs or Transformers can model temporal spending patterns and customer interactions more effectively. Reinforcement learning–based retention strategies may help optimize personalized offers in real time.e Additionally, developing profit-aware custom loss functions could further strengthen the alignment between model predictions and business objectives. Integrating external datasets — such as digital engagement metrics, credit bureau information or macroeconomic indicators — may also increase predictive accuracy. Automated hyperparameter tuning, real-time dashboards and more advanced interpretability methods beyond SHAP (e.g., counterfactual explanations) can further enhance the accuracy, transparency and practical deployment of customer churn prediction models.